
INTERNATIONAL GCSE **CHEMISTRY**

9202/2

Paper 2

Mark scheme

November 2019

Version: 1.0 Final



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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of error / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
01.1	neutron		1	AO1 3.1.2.d	A
01.2	electron		1	AO1 3.1.2.j	A
01.3	7		1	AO2 3.1.2.f	A
01.4	1 / one	allow alkali metals	1	AO1 3.1.3.b	E
01.5	+1		1	AO2 3.2.1.c	A
01.6	giant		1	AO1 3.2.2.a	E
01.7	high melting point		1	AO1 3.2.2.a	A
Total			7		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
02.1	C (because C is) resistant to corrosion (and C) is stronger than D	ignore references to density allow (because) C has a high (relative) strength if no other mark awarded allow 1 mark for (because) B is strong(est) or (because) D is resistant to corrosion	1	AO3 3.3.1.b	E
			1		
			1		
02.2	carbon		1	AO1 3.3.1.b	A
02.3	different composition	allow contains different amounts of the other elements	1	AO2 3.3.1.b	E
		or allow contains additional element(s)			
02.4	$2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$		1	AO2 3.3.1.1.d	A
02.5	oxygen is removed / lost (from iron oxide)	allow gain of electrons (by Fe(III) ions)	1	AO2 3.3.1.1.d 3.8.4.a	E
02.6	CaCO_3		1	AO2 3.3.1.2.a	E
	CO_2		1		

Total			9
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Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
03.1	any two from: • volume of sodium thiosulfate • volume of acid • concentration of acid • temperature (of solutions)		2	AO1 3.8.1.e	E
03.2	carry out in fume cupboard / well ventilated room		1	AO3 3.8.1.e	E
03.3	$(M_r \text{ sulfur dioxide} = 32 + 16 + 16 =) 64$ $(\text{moles} = 0.20 \times 0.025 =) 0.005 \text{ mol}$ $(\text{mass} = 0.005 \times 64 =) 0.32 \text{ (g)}$	an answer of 0.32 (g) scores 3 marks allow correct use of incorrectly calculated value of M_r allow correct use of incorrectly calculated value of moles of SO_2	1 1 1	AO2 3.6.1.d 3.6.2.a 3.6.4.a	E
03.4	extrapolation on graph value of time (s) read at 0.04 mol/dm ³ from extrapolated line	allow tolerance of $\pm \frac{1}{2}$ a small square	1 1	AO3 3.8.1.e	E
03.5	Result 0.12 (mol/dm ³) Reason (any one from): • a timing error was made • used 0.094 (mol/dm³) instead of 0.12 (mol/dm³)		1 1	AO3 3.8.1.e	E
03.6	as concentration increases, time taken decreases at a decreasing rate		1 1	AO3 3.8.1.e	E

03.7	reproducible		1	AO1 3.8.1.e	A
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03.8	$\frac{0.15}{20}$		1	AO2 3.8.1.a	E
	0.0075		1		
	g/s		1		

Total			16		
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Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
04.1	$1500 \times 7.20 = 10\,800 \text{ (g)}$ (=) 10.8 (kg)	an answer of 10.8 (kg) scores 2 marks	1	AO2 3.1.2.k	E
			1		
04.2	(trend is) as relative atomic mass increases, density increases		1	AO3 3.1.2.k	E
	this applies for chromium, copper and iron		1		
	(but) zinc does not follow the trend		1		
	(as) zinc has lowest density and highest mass		1		
04.3	(zinc + hydrochloric acid →)	allow any order		AO2 3.5.2.a 3.6.1.a	E
	zinc chloride		1		
	hydrogen		1		
04.4	Level 3: The design/plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.		5–6	AO4 3.3.1.1.a	E
	Level 2: The design/plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.		3–4		
	Level 1: The design/plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.		1–2		
	No relevant content		0		
	Indicative content Method - weigh metal - same mass / moles of each metal - add metal to suitable container - measure set volume of acid - add acid to container				

	- collect gas - measure volume of gas - suitable apparatus: e.g. conical flask connected to gas syringe or inverted measuring cylinder over water. - measure time take to collect set volume of gas or - measure volume of gas collected in a set time - repeat with other metals Control Variables - same surface area of metal - same concentration of acid - same temperature of acid - same mass of metal			
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04.5	copper is less reactive than iron		1	AO1 AO2 3.3.2.h 3.3.1.1.a	E
	(so) doesn't react with dilute hydrochloric acid		1		
	(therefore) prevents dilute hydrochloric acid reaching iron	allow Cu cannot displace H ⁺ ions	1		

Total			17
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Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
05.1	oxidation		1	AO1 3.10.3.1.c	A
05.2	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O}-\text{H} \end{array} \\ \\ \text{H} \end{array} $		1	AO1 3.10.3.2.a	E
05.3	all points plotted correctly	allow tolerance of $\pm \frac{1}{2}$ square allow 1 mark for 3 or 4 points plotted correctly.	2	AO2 3.10.3.2.a	E
05.4	(ethanoic acid is a) weak acid (so) does not ionise completely (in water) (so) fewer H^+ ions are produced (so) ethanoic acid has a higher pH (than hydrochloric acid)	allow converse statements about hydrochloric acid	1 1 1 1	AO1 3.10.3.2.b 3.5.1.f	E
Total			8		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
06.1	number of electrons in the molecules increases		1	AO3 3.7.1.c	E
	bond energy decreases	if neither MP2 or MP3 are awarded allow 1 mark for bond energy increases then decreases	1		
	except for F – F / 157 (kJ/mol)		1		
06.2	damp blue litmus paper		1	AO1 3.4.2.e	E
	turns red		1		
	(then) turns white or is bleached		1		
06.3		Max 2 marks if no numerical values given		AO2 3.1.2.f 3.1.2.h	E
	both have 17 protons	allow both have same number of protons	1		
	both have 17 electrons	allow both have same number of electrons	1		
	³⁵ Cl has 18 neutrons and ³⁷ Cl has 20 neutrons	allow ³⁷ Cl has 2 more neutrons than ³⁵ Cl allow have different number of neutrons	1		

06.4	(bonds broken = $612 + (413 \times 4) + 243 =$ 2507 (bonds formed = $347 + (413 \times 4) + (346 \times 2) =$ 2691 2507 – 2691 –184 OR (alternative method ignoring the four unchanged C-H bonds) $(612 + 243 =)$ 855 (1) $(347 + (2 \times 346) =)$ 1039 (1) 855 – 1039 (1) $= -184$ (kJ/mol) (1)	an answer of –184 scores 4 marks an answer of 184 (kJ/mol) scores 3 marks an incorrect answer for one step does not prevent allocation of marks for subsequent steps	1 1 1 1	AO2 3.9.2.f	E
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06.5		(Cr)	(Cl)	correct formula without working scores 1 mark only	1	AO2 3.6.2.c	E
	$\left(\frac{\text{mass}}{A_r}\right)$	$\frac{2.6}{52}$	$\frac{5.3}{35.5}$				
	(moles)	0.05	0.15				
	(whole number ratio)	1	3				
	(formula)	CrCl ₃					
				allow correct whole number ratio from incorrect calculation of moles	1		
				allow multiples	1		

06.6	ammonia hydrochloric acid	answers may be in either order allow ammonium hydroxide	1 1	AO2 3.5.1.d 3.8.3.b	E
06.7	0.5 (cm ³)	allow values in range 0.45 - 0.54 (cm ³)	1	AO3 3.5.1.f	E
06.8	blue / purple		1	AO3 3.5.1.f	E
06.9	(reactants: NH ₄ ⁺ +) OH ⁻ (products: NH ₃ +) H ₂ O		1 1	AO2 3.5.1.g	E
Total			23		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
07.1	bubbling stops		1	AO4 3.4.1.c	E
07.2	to ensure all the (silver) carbonate reacts		1	AO4 3.4.1.c	E
07.3	wash solid / residue (with water) to remove impurities		1 1	AO4 3.4.1.c	E
07.4	(Y is) calcium sulfate or magnesium sulfate.	allow 1 mark for Y contains calcium or magnesium (ions) or Y contains sulfate (ions)	2	AO3 3.4.3.b 3.4.3.f	E
07.5	$(M_r(\text{Z}(\text{OH})_2) =) A_r(\text{Z}) + 34$ $A_r(\text{Z}) = 0.8 \times (A_r(\text{Z}) + 34)$ $(A_r(\text{Z}) =) \frac{27.2}{0.2}$ $A_r(\text{Z}) = 136$ alternative approach: $(M_r(\text{OH})_2 =) 34 (1)$ $(20\% \text{ of } M_r = 34, \text{ so } M_r \text{ Z}(\text{OH})_2)$ $= 5 \times 34 (1)$ $= 170 (1)$ $(170 - 34 =) 136 (1)$	an answer of 136 scores 4 marks	1 1 1 1	AO2 3.6.2.a	E
Total			10		